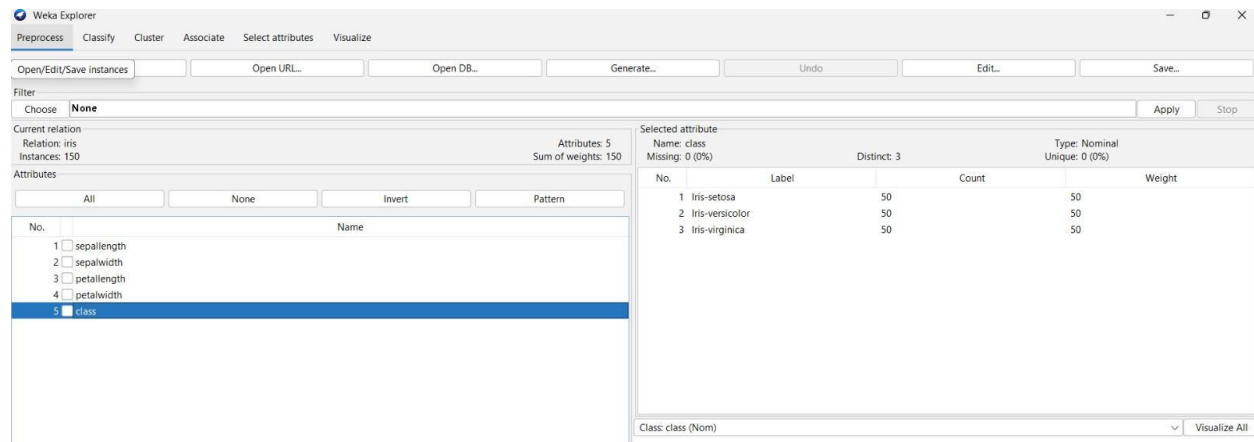


EXP21 DATA PREPROCESSING AND ANALYSIS FOR DATASET USING WEKA



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open/Edit/Save instances Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: iris
Relation: iris
Instances: 150
Attributes: 5
Sum of weights: 150

Attributes: All None Invert Pattern

No.	Name
1	sepal.length
2	sepal.width
3	petal.length
4	petal.width
5	class

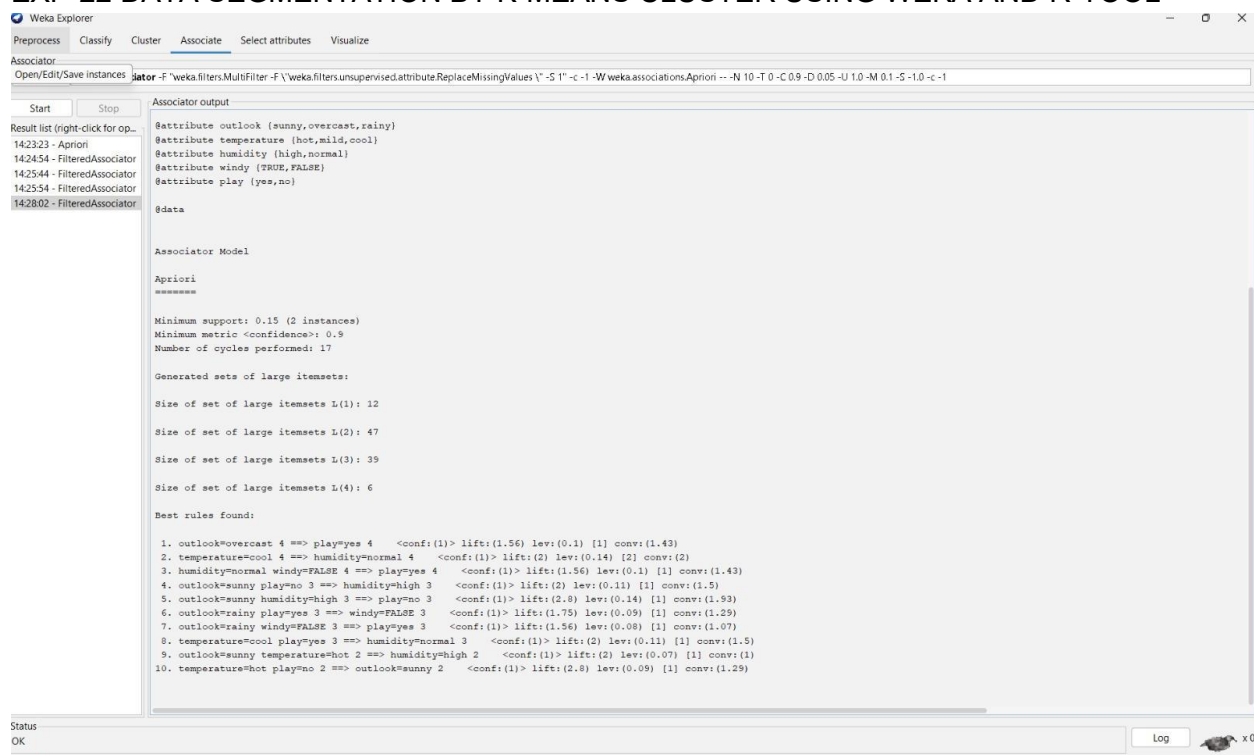
Selected attribute

Name: class
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	Iris-setosa	50	50
2	Iris-versicolor	50	50
3	Iris-virginica	50	50

Class: class (Nom) Visualize All

EXP-22 DATA SEGMENTATION BY K-MEANS CLUSTER USING WEKA AND R-TOOL



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Open/Edit/Save instances **Start** **Stop**

Result list (right-click for op...): 142323 - Apriori, 142454 - FilteredAssociator, 142544 - FilteredAssociator, 142554 - FilteredAssociator, 142802 - FilteredAssociator

Associator output

```
@attribute outlook {sunny,overcast,rainy}
@attribute temperature {hot,mild,cool}
@attribute humidity {high,normal}
@attribute windy {TRUE,FALSE}
@attribute play {yes,no}

@data

Associator Model

Apriori
*****

Minimum support: 0.15 (2 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 17

Generated sets of large itemsets:

Size of set of large itemsets L(1): 12
Size of set of large itemsets L(2): 47
Size of set of large itemsets L(3): 35
Size of set of large itemsets L(4): 6

Best rules found:

1. outlook=overcast 4 ==> play=yes 4 <conf: (1)> lift: (1.56) lev: (0.1) [1] conv: (1.43)
2. temperature=cool 4 ==> humidity=normal 4 <conf: (1)> lift: (2) lev: (0.14) [2] conv: (2)
3. humidity=normal windy=FALSE 4 ==> play=yes 4 <conf: (1)> lift: (1.56) lev: (0.1) [1] conv: (1.43)
4. outlook=sunny play=no 3 ==> humidity=high 3 <conf: (1)> lift: (2) lev: (0.11) [1] conv: (1.5)
5. outlook=sunny humidity=high 3 ==> play=no 3 <conf: (1)> lift: (2.8) lev: (0.14) [1] conv: (1.93)
6. outlook=rainy play=yes 3 ==> windy=FALSE 3 <conf: (1)> lift: (1.75) lev: (0.09) [1] conv: (1.29)
7. outlook=rainy windy=FALSE 3 ==> play=yes 3 <conf: (1)> lift: (1.56) lev: (0.08) [1] conv: (1.07)
8. temperature=cool play=yes 3 ==> humidity=normal 3 <conf: (1)> lift: (2) lev: (0.11) [2] conv: (1.5)
9. outlook=sunny temperature=hot 2 ==> humidity=high 2 <conf: (1)> lift: (2) lev: (0.07) [1] conv: (1)
10. temperature=hot play=no 2 ==> outlook=sunny 2 <conf: (1)> lift: (2.8) lev: (0.09) [1] conv: (1.29)
```

Status: OK Log x 0

EXPERIMENT- 23 DATA SEGMENTATION BY EXPECTATION MAXIMISATION ALGORITHM THROUGH WEKA

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

ssociator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1

Start Stop

result list (right-click for op...

14:23:23 - Apriori
14:24:54 - FilteredAssociator
14:25:44 - FilteredAssociator
14:25:54 - FilteredAssociator
14:28:02 - FilteredAssociator
14:29:28 - FPGrowth

Associator output

```

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation:     supermarket
Instances:    4627
Attributes:   217
              (list of attributes omitted)
=== Associator model (full training set) ===

FPGrowth found 16 rules (displaying top 10)

1. [fruit=t, frozen foods=t, biscuits=t, total=high]: 788 ==> [bread and cake=t]: 723 <conf:(0.92)> lift:(1.27) lev:(0.03) conv:(3.35)
2. [fruit=t, baking needs=t, biscuits=t, total=high]: 760 ==> [bread and cake=t]: 696 <conf:(0.92)> lift:(1.27) lev:(0.03) conv:(3.28)
3. [fruit=t, baking needs=t, frozen foods=t, total=high]: 770 ==> [bread and cake=t]: 705 <conf:(0.92)> lift:(1.27) lev:(0.03) conv:(3.27)
4. [fruit=t, vegetables=t, biscuits=t, total=high]: 815 ==> [bread and cake=t]: 746 <conf:(0.92)> lift:(1.27) lev:(0.03) conv:(3.26)
5. [fruit=t, party snack foods=t, total=high]: 854 ==> [bread and cake=t]: 779 <conf:(0.91)> lift:(1.27) lev:(0.04) conv:(3.15)
6. [vegetables=t, frozen foods=t, biscuits=t, total=high]: 797 ==> [bread and cake=t]: 725 <conf:(0.91)> lift:(1.26) lev:(0.03) conv:(3.06)
7. [vegetables=t, baking needs=t, biscuits=t, total=high]: 772 ==> [bread and cake=t]: 701 <conf:(0.91)> lift:(1.26) lev:(0.03) conv:(3.01)
8. [fruit=t, biscuits=t, total=high]: 954 ==> [bread and cake=t]: 866 <conf:(0.91)> lift:(1.26) lev:(0.04) conv:(3)
9. [fruit=t, vegetables=t, frozen foods=t, total=high]: 834 ==> [bread and cake=t]: 757 <conf:(0.91)> lift:(1.26) lev:(0.03) conv:(3)
10. [fruit=t, frozen foods=t, total=high]: 969 ==> [bread and cake=t]: 877 <conf:(0.91)> lift:(1.26) lev:(0.04) conv:(2.92)

```

EXPERIMENT- 24 DATA SEGMENTATION BY COBWEB – HIERARCHIAL CLUSTERING ALGORITHM USING WEKA TOOL

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **ZeroR** Classify instances

Options

Use training set

Supplied test set Set...

Cross-validation Folds 10

Percentage split % 66

More options...

Options

Start Stop

ult list (right-click for options)

31:16 - rules.ZeroR

Classifier output

ZeroR predicts class value: Iris-setosa

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	50	33.3333 %
Incorrectly Classified Instances	100	66.6667 %
Kappa statistic	0	
Mean absolute error	0.4444	
Root mean squared error	0.4714	
Relative absolute error	100	%
Root relative squared error	100	%
Total Number of Instances	150	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	1.000	0.333	1.000	0.500	?	0.500	0.333	Iris-set
	0.000	0.000	?	0.000	?	?	0.500	0.333	Iris-ver
	0.000	0.000	?	0.000	?	?	0.500	0.333	Iris-vir
Weighted Avg.	0.333	0.333	?	0.333	?	?	0.500	0.333	

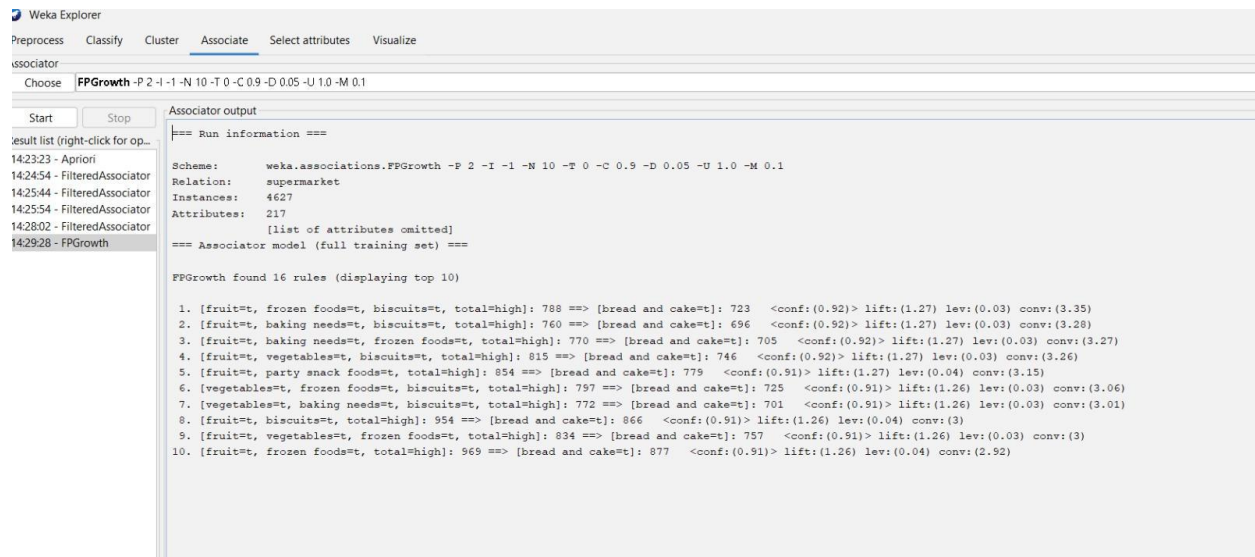
=== Confusion Matrix ===

```

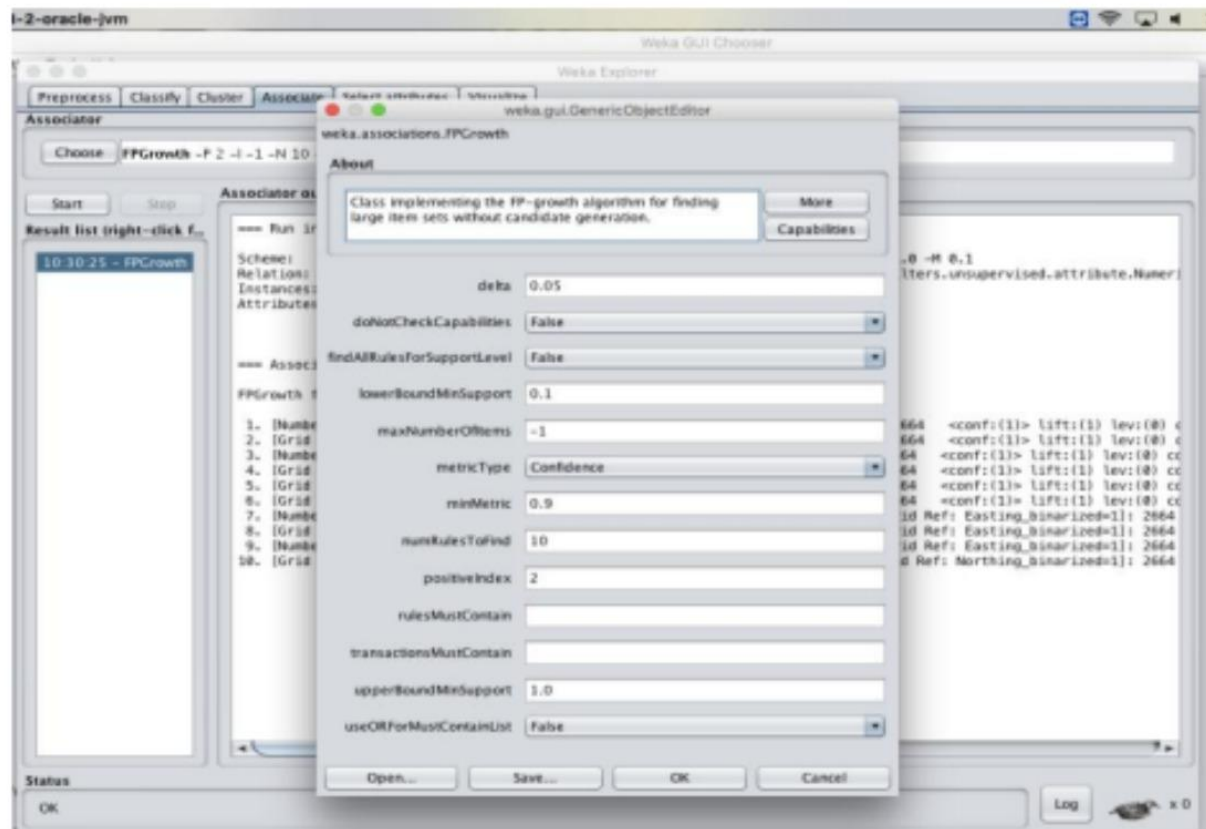
a b c <-- classified as
50 0 0 | a = Iris-setosa
50 0 0 | b = Iris-versicolor
50 0 0 | c = Iris-virginica

```

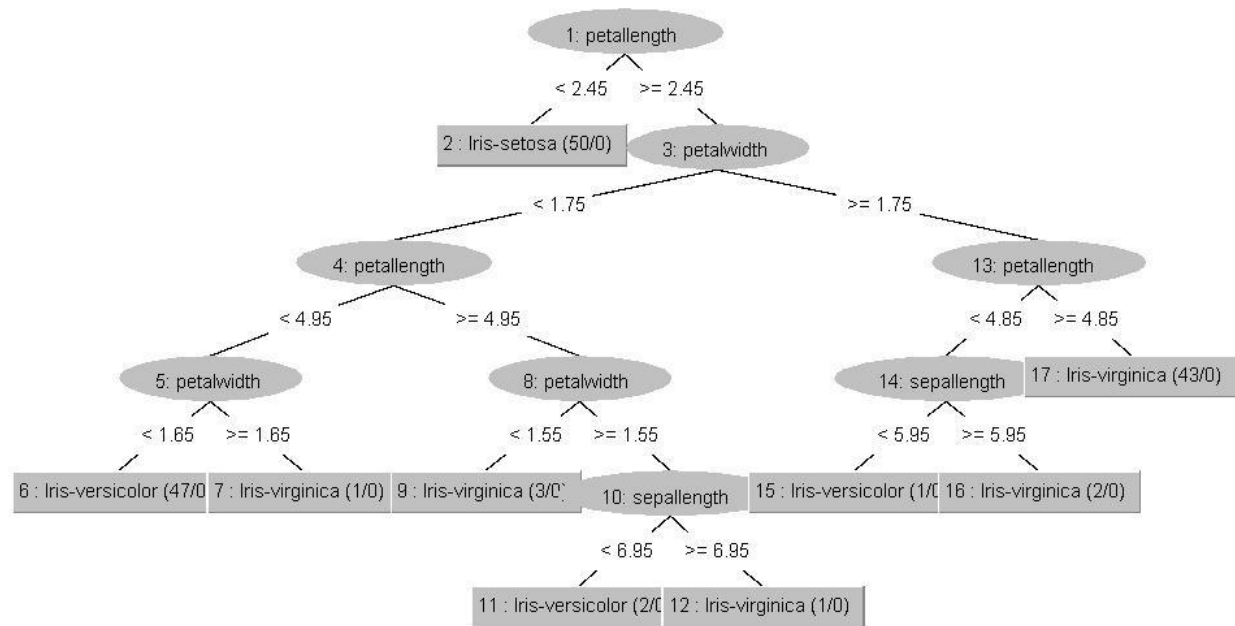
EXPERIMENT- 25 FREQUENT PATTERN MINING USING ASSOCIATION RULE THROUGH WEKA AND R -TOOLS



EXPERIMENT- 26 FREQUENT PATTERN MINING USING FP GROWTH THROUGH WEKA TOOL



EXPERIMENT- 27 PREDICTION OF CATEGORICAL DATA USING DECISION TREE ALGORITHM THROUGH WEKA



EXPERIMENT- 28 PREDICTION OF CATEGORICAL DATA USING SMO ALGORITHM THROUGH WEKA

Process: **Classify** | Cluster | Associate | Select attributes | Visualize

Classifier: Choose **ZeroR** | Classify instances

Options:

- Use training set:
- Supplied test set:
- Cross-validation: Folds
- Percentage split: %
- More options...

Output:

Classifier output

ZeroR predicts class value: Iris-setosa

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	50	33.3333 %
Incorrectly Classified Instances	100	66.6667 %
Kappa statistic	0	
Mean absolute error	0.4444	
Root mean squared error	0.4714	
Relative absolute error	100	%
Root relative squared error	100	%
Total Number of Instances	150	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	1.000	0.333	1.000	0.500	?	0.500	0.333	Iris-set
	0.000	0.000	?	0.000	?	?	0.500	0.333	Iris-ver
	0.000	0.000	?	0.000	?	?	0.500	0.333	Iris-vir
Weighted Avg.	0.333	0.333	?	0.333	?	?	0.500	0.333	

=== Confusion Matrix ===

```

a b c <-- classified as
50 0 0 | a = Iris-setosa
50 0 0 | b = Iris-versicolor
50 0 0 | c = Iris-virginica

```

EXPERIMENT- 29 EVALUATING ACCURACY OF THE CLASSIFIERS

```

lassifier output
Intercept          8.1743          42.637

Odds Ratios...
      Class
Variable  Iris-setosa  Iris-versicolor
=====
sepalength  2954196659.8836      11.7653
sepalwidth   96.0426      797.0304
petalength      0      0.0001
petalwidth      0      0

Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      144          96      %
Incorrectly Classified Instances      6           4      %
kappa statistic                    0.94
Mean absolute error                 0.0287
Root mean squared error             0.1424
Relative absolute error              6.456 %
Root relative squared error         30.2139 %
Total Number of Instances          150

=== Detailed Accuracy By Class ===
      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000    Iris-setosa
      0.920    0.020    0.958     0.920    0.939     0.910    0.972    0.934    Iris-versicolor
      0.960    0.040    0.923     0.960    0.941     0.911    0.972    0.934    Iris-virginica
Weighted Avg.  0.960    0.020    0.960     0.960    0.960     0.940    0.981    0.956

=== Confusion Matrix ===
  a  b  c  <-- classified as
50  0  0 |  a = Iris-setosa
 0 46  4 |  b = Iris-versicolor
 0  2 48 |  c = Iris-virginica

```

EXPERIMENT- 30 DESCRIPTION NUMERICAL PREDICTION ANALYSIS USING LINEAR REGRESSION THROUGH WEKA